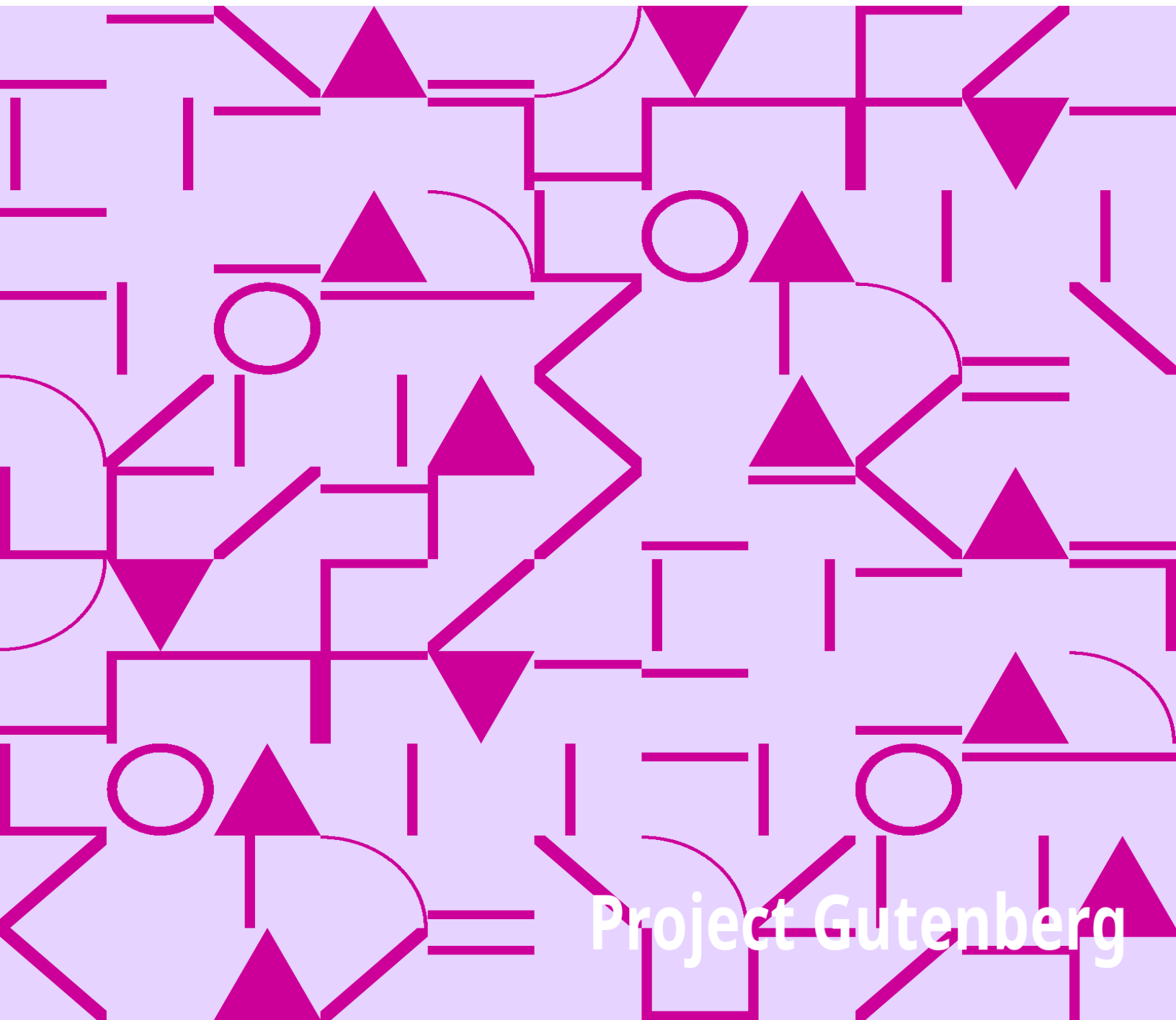


# Remarks on the Proposed Railway Between Birmingham and London

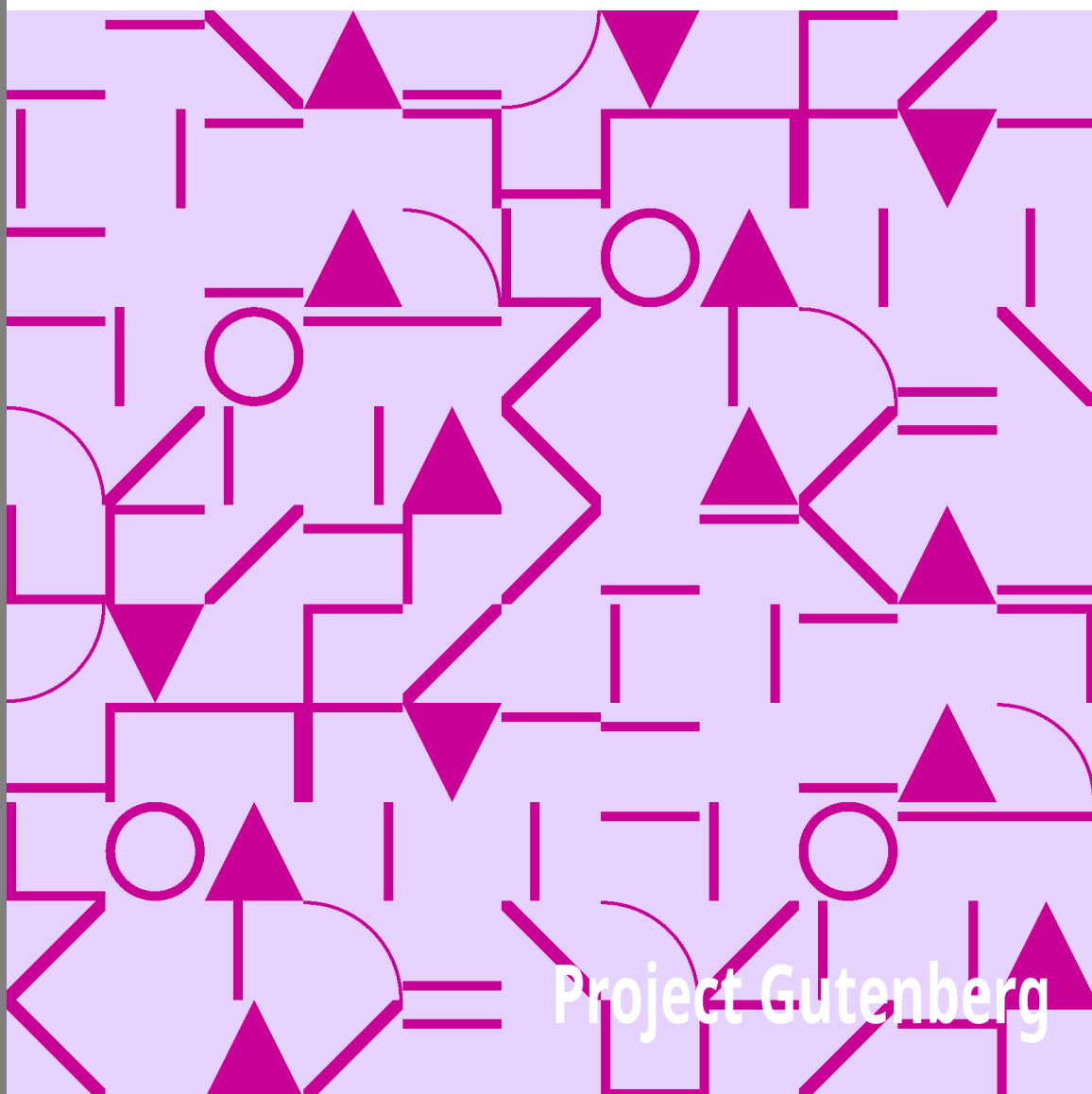
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THE PROPOSED RAILWAY BETWEEN BIRMINGHAM AND  
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**REMARKS  
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THE PROPOSED  
RAILWAY  
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## REMARKS, &c.

WITHOUT minutely inquiring into the origin of the different modes of conveyance at present existing in this country and others for passengers and goods, I shall content myself with asking, Why were canals first established? and What was the great benefit arising from them, which caused so much as fifteen hundred miles in extent to be executed in less than a quarter of a century, at a cost of nearly twenty millions of money, and for the most part during a time of war, when the highest rate of taxation prevailed?

Previously to the establishment of canals,—roads, waggons and horses were the means employed for the conveyance of goods; and the speed which they accomplished was greater than the average speed afterwards obtained by canals. But notwithstanding this advantage, it was found, that to carry a ton weight of grain, coal, or merchandize one hundred miles cost upwards of 6*l*. Hence materials and goods, whose weight bore a great proportion to their value, could only be conveyed a few miles from the spot where they were raised or manufactured; and thus, sources of wealth that have since been highly productive and profitable were shut up and useless. Baron Dupin in his excellent work on the Commercial Power of Great Britain, published 1825, states, “Up to 1756 England had not a single line of artificial navigation; she possessed for communication by land, only a small number of roads injudiciously cut and ill kept up. Of a sudden an individual conceives the idea to profit by the general impulsion which industry had received, by cutting a canal to carry to Manchester the product of his mines. Shortly afterwards, a town which thrives, and of which the exuberant wealth seeks everywhere productive outlets,—Liverpool,—aspires to still higher designs. She is the first to form and realize the project of opening a navigable channel between the Irish Sea and the German

Ocean. Other channels even more extended are opened by degrees: thus, within the short space of half a century, a double row of canals is formed, both for great and small navigation, for the purpose of uniting together opposite seas; basins separated by numberless chains of hills and mountains; opulent ports; industrious towns; fertile plains; and inexhaustible mines;—and this presents a development of more than a thousand leagues in length, upon a portion of territory not equal to one-fourth of France! The roads which already existed are enlarged, are reconstructed with more art, and kept up with more care. New channels are thrown open to commerce, and a system of roads is now being formed, of which the total length is at present (1825) more than 46,000 leagues in England alone. Thanks to these works, at this moment, in the three kingdoms, 22,300 merchant-vessels, manned by 160,000 men, and capable of carrying two millions of tons of merchandize, are scarcely sufficient for the exportation of the superfluity of interior circulation, for the trade along the coast, and for the importation of those foreign products necessary to keep up a circulation so immense.”

As further instances of the effect produced by the same causes,—in 1740, before the establishment of canals, the iron manufactured in England and Scotland employed 59 furnaces, which produced annually 17,000 tons. In 1827 there were upwards of 280 furnaces, with an annual produce of 690,000 tons; during the intervening period canals were cut, connecting the iron districts with large towns and the ports. In 1750 there was but one smelting furnace in Staffordshire, making less than 2,000 tons of iron per annum. In 1827 there were 97 furnaces in that district only, making 216,000 tons per annum. The population of Staffordshire in 1750 was 160,000; it is now upwards of 350,000. In England in 1750 it was 6,017,000; it is now upwards of 13,000,000.

The total amount of the exports in 1750 was 7,772,039*l.*; in 1824 it was 56,234,663*l.* In 1760 the number of ships assessed in Liverpool was 1,245; in 1824, it was 10,001. These statements are sufficient to show the changes consequent upon the introduction of canals. It was found, that one horse upon a canal could convey twenty times what he could upon a road; and that what formerly cost six pounds to send a hundred miles, could by canals be sent for little more than one pound. Such is the traffic upon the Birmingham Canal alone, that at some periods there is a weight of goods

and materials brought by 150 horses and boats, in one day, which by the roads would require 3,000 horses and 1,000 waggons. We find again that the population has more than doubled itself in three principal inland towns, viz. Birmingham, Manchester, and Leeds, since the establishment of canals.

But notwithstanding the advantages that have attended upon the introduction of canals, there are limits within which their utility is confined, and, as regards despatch, much confined. The canals as they are now constructed are adapted only to horse power, and are subject to the inconvenience of that slowness of travelling which arises from the great increased resistance of fluids to bodies moving in them, with only a very slight increase of velocity. Two horses may take a loaded boat of twenty-five tons at the rate of four miles per hour; but to obtain a velocity of twelve miles per hour, it would require twenty-seven horses. It is found that with a velocity of six miles per hour so great a surge or swell is produced in the water as to hazard the sinking of any boats that are passing.

The stoppages arising at the locks is very considerable. In the canals between Birmingham and London, every means are used to effect despatch; but still the quickest passage for the fly-boats is sixty hours. The distance is 153 miles, and there are 142 locks; nearly one-third of the time is lost in passing them; and while this is being done, one horse and four men are comparatively idle; the expenses of wages and keep however are going on. These expenses are incurred more particularly by the haulage; but in addition to them the tonnages are very high, and of necessity so, since the repairs of so many locks, cleaning and repairing canal, and above all, the raising of water to the summit-levels by steam-engines, must incur a great expense. The cost of this last operation may be guessed at, when it is known that for every boat that passes from London to Birmingham, a body of water of 120 tons weight has to pass through a difference of level of 1,140 feet. And yet in dry summers, notwithstanding the pains thus bestowed, the boats are frequently detained, for want of water, twelve, eighteen, and twenty-four hours in one trip.

Much, then, as the canals surpass the common roads, it appears that much remains to be done, if their peculiar disadvantages can be got rid of.

The economy of steam compared with horse power is too well established to need many comments. The expense of working a twenty-horse power steam-engine is known to be less than one-sixth of the cost of twenty horses and men to attend them. I appeal to the proprietors of the thousands of steam-engines now in use for the truth of this statement. Does not, then, a question at once arise, Whether it is not possible to construct a road of some kind upon which this great and cheap power may be made to act, instead of horses, and with as great a degree of profit? We rejoice that the question has suggested itself, and that it has received an answer. The application of steam to the purpose of locomotion has been proved upon a rail-road between Manchester and Liverpool. Moreover, the economy of steam power when applied to rapid progression is found to be much greater than when it is employed to supersede the horse-mill. In this the animal is not driven beyond his speed, but is allowed to move at a rate of two miles and a half per hour,—a speed which he can continue to perform for eight or ten hours per day. A horse is found to perform the greatest quantum of work when moving at this rate, and to be employed most economically. But even compared with this, we have seen that an equal effect may be produced by steam power at one-sixth the cost. With how much greater advantage and profit, then, can steam be applied, where it is made to supersede the use of horses in rapid motion, of eight or ten miles per hour! at which work it is known they can continue but one hour per day, and even then they do not live to half the age of the horse employed at the slower speed. It is a fact, that horses employed in the fast coaches and for the fly-boats require renewing every four years. Can steam power, then, be used to produce this rapid motion? One hundred thousand persons that have passed from Liverpool to Manchester in less than two hours since September last, can answer this question in the affirmative.

Another great advantage of the railway over the common road is the much greater safety that it offers. The above number of passengers have been conveyed with scarcely an accident. Is there a line of turnpike-road in the kingdom that can make the same boast, where an equal number of passengers have travelled? The general causes of accident upon a common road, such for instance as the horses taking fright, the coachman losing command over them, the coach running against some obstacle in the road, or upsetting, cannot happen upon a railway. The engine can exert no

*additional* force without the consent of the man who superintends it; one carriage cannot meet another, as there is a separate line for carriages travelling in the opposite direction; and even if a wheel breaks, the carriage is so low that it would not upset. The only objection that is started on this head, is the bursting of the boiler: this is now constructed with a number of small tubes, through which the fire passes, and which are all inclosed in a strong iron case; so that if one of these small tubes burst, it immediately acts as a safety-valve, discharging the steam of the boiler into the chimney, and removing the apprehended danger.

Here then we have safety and despatch, and we may next consider economy in travelling. In this respect the public have also found an advantage. The charges by the Liverpool and Manchester railway are not half what the coaches formerly charged, the fares being but 3s. 6*d.* and 5s. each. And the effect produced is, that the average number of passengers by the railway is upwards of 1,000 per day. The average number by the coaches the preceding year was only about 500 per day, paying 7s. outside and 12s. inside. Goods are now taken in three hours, and at a charge of 11s. per ton; they were before thirty-six hours on the water-conveyance, at a charge of 15s. per ton.

Having then referred to what has been done upon a line of railway from Liverpool to Manchester, with a profit to the proprietors and great benefit to the public, let us inquire whether there is the same want of an improved conveyance from Birmingham to London. We will first consider the amount that is now paid, and the time that is now required for conveying both goods and passengers. The amount paid for the conveyance of passengers and goods between Birmingham and London is upwards of 800,000*l.* per annum; more than half of which is comparatively at a high rate, as will be seen by the following statements.

The charges by coaches for passengers are 42s. inside, and 21s. outside, and for parcels 1*d.* per pound. By waggon, the charge is 5s. per cwt. By fly-boat, for packages and general merchandise, 60s. per ton; and for iron-castings in boat-loads, 32s.; for pig-iron, 25s. per ton. Thus the coach charge for luggage, at ten miles per hour, is 1s. 9*d.* per ton per mile; waggon charge, for a rate of three miles and a half per hour, is 10¾*d.* per ton per mile; and goods by fly-boat, at two miles per hour, 6½*d.* per ton per mile;

and the lowest rate for pig-iron  $2\frac{1}{2}d.$  per ton per mile. These charges are calculated upon a distance of 110 miles.

|                                                                                                                      | £        |
|----------------------------------------------------------------------------------------------------------------------|----------|
| Twenty-two coaches pass daily each way, which, at 15 <i>l.</i> per coach (the expense of working it), cost per annum | 227,600  |
| Goods per van and waggon                                                                                             | 36,300   |
| Goods per canal, paying from 40 <i>s.</i> to 60 <i>s.</i> per ton, about 180 tons per day                            | 146,000  |
| Making a total annual amount of                                                                                      | £409,900 |

for only a portion of the business; as in the above statement none of the coaching or posting that falls upon the Holyhead road, at Coventry, Northampton, and many of the towns nearer London, is taken into the account. The above amount is calculated from what passes through Birmingham only, and that exclusive of posting.

In addition to the above traffic, there are upwards of 2,000 tons that pass daily through Birmingham, besides an equal amount upon the Grand Junction Canal. There is also the whole of the business that arrives at the different towns upon the line; and it is ascertained that the passengers by stage-coaches only that pass through the towns near London, are upwards of four times the number that pass through Birmingham. But if it appears that a considerable profit would arise from conveying but a portion of the business by a railway, it will be much more satisfactory than if it were dependent upon the whole for an adequate return. And yet, as it appears that passengers are taken by this mode of conveyance in half the time in which they can be taken by any other, and at half the cost, and that goods are conveyed in one-sixth of the time and at a lower charge, the great probability is that a considerable portion will immediately be sent by it.

We will now inquire what would be the cost of conveying by a railway between Birmingham and London, the passengers and goods which are now paying by the roads and canals 409,900*l.* per annum.

It has been ascertained upon the Liverpool and Manchester Railway, that the whole expenses of one engine, capable of conveying twenty-five tons of goods 105 miles per day, at a rate of from ten to fifteen miles per hour, including fuel, attendance, repairs, oil, grease, &c., is 500*l.* per annum. Its eight waggons (the number required for 25 tons) at 30*l.* per annum each, involve a yearly expense of 240*l.* This, with the 500*l.*, makes a total cost of 740*l.* We will take it at 800*l.* per annum, which gives the following cost, on the supposition that each engine takes but twenty tons:

|                                                                                                                                                                                                                                                                                                                                                                                                                | £       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| For conveying 180 tons per day it would require ten engines, which, with the waggons, cost 800 <i>l.</i> each per annum                                                                                                                                                                                                                                                                                        | 8,000   |
| Supposing the present coaches to carry two tons each of passengers and parcels, or a whole weight daily of forty-four tons, which we will suppose requires four engines; and taking the repairs of the coaches to be double that of the waggons, or 60 <i>l.</i> per annum each; it will make a total amount of 980 <i>l.</i> , or say 1000 <i>l.</i> per annum, for the four engines and four sets of coaches | 4,000   |
| Goods per van and waggon twenty-six tons per day, which we will also suppose require two engines                                                                                                                                                                                                                                                                                                               | 1,600   |
|                                                                                                                                                                                                                                                                                                                                                                                                                | £13,600 |
| Supposing they are but two-thirds loaded, we will add to the expenses                                                                                                                                                                                                                                                                                                                                          | 4,400   |
|                                                                                                                                                                                                                                                                                                                                                                                                                | £18,000 |

We then have a total cost of only 18,000*l.*, exclusive of railway dues, for conveying that by steam power which the public are now paying 409,900*l.* for conveying by horses upon roads and canals in one district only. This, as before stated, is only a portion of the business. Supposing 2,000 tons of goods to be conveyed daily at 20*s.* per ton, which is less than two-thirds of the present average charge; and taking 300 days per annum, we have 600,000*l.*; that might be taken by 200 engines, which with the waggons would cost, at the rate of 800*l.* each, 160,000*l.* per annum. Nothing is calculated here for the conveyance of cattle, which may be reckoned upon,

when it is known that the present cost of conveying an ox from Daventry or Leicester to London, including the loss of weight by driving, is 30s.; and when it is also known that a weight equal to an ox may be taken that distance in five hours for 7s. Sheep may also be taken at a proportionately low rate. Another great source of income and profit will arise from the conveyance of carriages and horses, as both will be taken with ease and safety in vehicles constructed for the purpose.

The question we will next consider is the amount of capital required for making such a road. The London and Birmingham Railway is stated at 3,000,000*l*. This is taken from the most accurate estimates; but for the satisfaction of those who would like to have corroborative statements, we will compare it with the cost of the Liverpool and Manchester Railway. This undertaking, it is stated, may cost, including its carrying department and engines, carriages, &c., 1,000,000*l*.; and the railway is thirty-two miles in length. (For particulars see Appendix, No. I.)

The London and Birmingham Railway will be 105 miles in length; and therefore, constructed on the same scale, and its cost calculated at the same rate per mile, it would require, including every expense, rather more than 3,000,000*l*. But it must be recollected that materials and wages are at least twenty per cent. lower now than at the time the Liverpool and Manchester Railway was constructing. The cost of the rails for the Liverpool and Manchester Railway was 12*l*. 10s. per ton; they may now be purchased for 8*l*. 10s. per ton. The considerable reduction that has taken place in the wages of labour must be taken into account, particularly as three-fourths of the whole cost will be in an expenditure of this nature upon the line. The great reduction that has taken place in all kinds of ironwork will tend to reduce the cost of the engines, and the machinery required in the construction of the road.

It must not be forgotten, that the Liverpool and Manchester being the first line of the same magnitude that had ever been constructed, many expenses occurred, which are always attendant upon works of a new kind, and which are much reduced when the same kind of work has to be executed again. It was formerly the opinion of those who are best acquainted with these subjects, that railways might be constructed at as little expense as canals: and when we consider that a canal requires a perfect level; occupies an

equal width of land with a railway; must have its embankments made of materials impervious to water;—and when, again, it is known that an expensive lock is required at almost every mile, as well as numerous drains and reservoirs to collect the water from the surrounding country, and powerful engines to raise the water from the lower to the summit levels;—it does appear a natural supposition, that such a work must require a greater outlay than a road, which, to a degree, has the inclination suited to the level of the country over which it passes; whose embankments, instead of being washed by a river, have only to carry two or four bars of iron; and, moreover, in whose banks no injury from leakage can arise to the country lying below it; which invariably happens to land at the foot of canal embankments.

That funds will be provided for the London and Birmingham Railway far beyond any amount expended in canals, will be seen by the following statement:—In 1825, it is stated, there were in England ninety-seven canals, the total extent of which was 2,471 miles, and the cost of which was 30,000,000*l.*; giving an average cost of about 12,500*l.* per mile. The railway from London to Birmingham will be 105 miles in length, and capital to the amount of 3,000,000*l.* will be provided for it: this gives an average of upwards of 28,000*l.* per mile. That the land in this direction is more favourable to works of this kind than that in many other parts of the country, is known by the canals that run in this district costing less per mile than the average rate of others. But even if the Railway costs twice or three times as much as a canal, the advantages that it has over a canal in its amount of traffic, and the economy attending it, are in a much greater proportion than the increase of its cost.

There are very few canals of any extent that can pass more than two hundred tons of goods per hour. The locks in general admit of but one boat passing at a time, carrying from twenty to twenty-five tons. Where the greatest despatch is used, the average is not more than eight boats passed per hour. Here at once is a limit to the despatch of canal conveyance. Eight boats per hour, at an average load of twenty tons, gives only 160 tons per hour as the greatest quantity that can be passed. The average load of a fly-boat is but sixteen tons.

Let us now inquire what might be conveyed along a single line of railway, on the supposition that a speed of ten miles per hour only may be performed upon it. The number of yards in a mile (1,760) multiplied by the speed, gives 17,600. A carriage for three tons of goods occupies a space of four yards and a half; but we will suppose a space of six yards is required, which gives two yards for each ton. Then 17,600 (the number of yards per mile multiplied by the velocity) divided by 2 yards, gives 8,800 tons per hour, on the supposition that the carriages moved in a continuous train; but with a space between each train of carriages equal in length to the train, we have half that amount, or 4,400 tons, that might if necessary be conveyed upon a single line of railway: or, what requires twenty-four hours to pass through the canal locks at Birmingham, might be sent along a railway in one hour. Thus it appears, that canals are limited, in respect of the quantity of goods that can be conveyed upon them, to less than 200 tons per hour; their greatest average speed is less than three miles per hour, and for the slow boats one mile and a half per hour; the general expense of haulage is 1*d.* per ton per mile; they are subject to stoppages averaging five weeks per annum, from frost, drought, and repairs. A railway, on the contrary, may convey 4,000 tons per hour, at a velocity of fifteen miles: steam propelling power costs only one farthing per ton per mile, exclusive of the waggons; and the stoppage that would arise from the breaking of a rail, would not be more than one of half an hour.

Does not this show, then, that an infinitely superior conveyance is now offered to the public, both for themselves and their goods? An instance of the support which the public give to quick conveyance, is afforded by the facts, that in 1800 there were seven coaches from Birmingham to London, and the average time was eighteen and twenty hours; that at present there are twenty-two coaches, and the average time is twelve hours.

Last year the number of passengers from Liverpool to Manchester was 500 per day, and the time occupied by the journey four hours. Since the opening of the Railway in September last, the average number has been nearly 1,000 per day, and the time two hours. The increase of passing between the towns that have had steam communication has been in a much greater proportion than the above. Between London and Margate, Dover and Calais, Liverpool and Dublin, Liverpool and Greenock, Stockton and Darlington, the passing has increased nearly ten-fold since the

establishment of such improved conveyance. It is calculated that the whole number of passengers by steam-boats is one million and a quarter per annum. Can it then be expected that a steam communication from Birmingham to London will be an exception? for there is not a line to be found of equal extent in the kingdom on which the population is so great, or on which the commercial and agricultural transactions are so important. As the utility of railroads may be considered established by the one now in operation, it must be evident how desirable it is to connect Liverpool, Manchester, and Birmingham, and ultimately other towns in the North, with the Metropolis. Supposing a railroad connecting these important towns to exist, which might be considered as the chief line of communication from the North to the South, there is a reasonable hope, that shortly other roads would be proposed and executed, connecting the remaining influential towns. An expeditious, cheap, and secure conveyance would thus be established throughout the country; so that in all probability, the business that would be brought upon the main line by these side channels would increase the whole traffic beyond what can now be calculated upon.

It may be objected, that railroads would throw out of employment a great number of people, who are now actively engaged, as coachmen, guards, horse-keepers, boatmen, waggoners, &c.: but as all improvements which tend to reduce the price of travelling and quick communication have led to a different result, we may infer that the number of cross coaches and short conveyances of all kinds that would be established to bring up passengers and goods to the railway, would at once give employment to these men. We may take as a proof the present posting and travelling upon the road from London to Dover, which was greater in 1829 than ever it was known to be before, although in the same year upwards of 1,000 passengers were conveyed weekly from London to Calais by steam. If indeed there are some persons that will be thrown out of employment for a short time, we must set against this the vast increase of labour that will be given to mechanics and others employed in manufacturing the engines and machines used on railways; for if these works go forward, there will be immediate employment for tens of thousands of labourers, and constant occupation in the workshop for double the number of hands that for a time may be thrown out of work upon the roads or canals. And I would ask, Which is the more

important member of the community, the ingenious mechanic or the wandering boatman?

Let us again consider, that we are generating a new power, with the consumption only of a mineral drawn from the bowels of the earth, while we are saving the surface land to produce that sustenance which our increasing population requires. Every horse that is dispensed with, saves the produce that would support six men; and it is calculated that one-third of the grain consumed, is by horses <sup>[21]</sup>. The immense surface that is now required for the support of cattle may be conjectured, when it is known that in England and Wales alone we have upwards of thirty millions of acres in tillage, and that one acre may grow as much corn as is consumed by three men in a year. And yet we are in the habit of importing a considerable quantity of grain!

It is unnecessary to establish by reasoning what is borne out by facts; and I refer again to the railroad between Liverpool and Manchester, on which goods and merchandise are now taken in one-tenth of the time and at two-thirds of the former cost, and passengers in half the time and at half the former charges by coach. May not the same support be looked for between London, and Birmingham, when the same advantages are offered? May not also the support of Government be expected, when its sanction only is asked to establish such a communication from one side of the kingdom to the other, not a farthing to defray expenses being required from it; at the same time that the capability is offered it, of sending despatches from London to Liverpool in seven hours, or of transporting twenty thousand troops the same distance within twelve hours, in case of need?

Though the capital required is considerable, yet I think the public may be trusted in seeking out their own means of investment; they are in fact only carrying *that* principle to the establishment of an improved conveyance at the cost of an old one, (an improvement in which many of them are deeply interested,) upon which in their manufactories they constantly act, when a machine that costs 100*l.* is supplanted by one that may cost 500*l.*, but which is found to perform ten times the work. Why, I ask, may not 3,000 individuals associate themselves together to apply to a public undertaking the mechanical power which they have found so advantageous in their private establishments? Why not extend to the purpose of locomotion the

power which has raised the manufactures of Great Britain and Ireland to their present high state of improvement? I would ask those who oppose its extension to this new purpose, whether the mechanical agencies of this kingdom, now amounting to the physical force of ten times the population of England, would ever have been called into action if the application of known powers and principles to fresh purposes and uses had not at all times met with the fullest support of the Legislature and the Public?

I would ask again, Whether the steam power that has been generated within the last fifteen years, for propelling vessels exceeding the whole physical force of the British navy, would have had birth, had the principle been acted upon, of things remaining as they are?

*March 4th, 1831.*

# APPENDIX.

## No. I.

*General Abstract of Expenditure of the Liverpool and Manchester Railway, to the 31st of May 1830, [from Mr. Booth's Pamphlet.]*

|                                                                                                                                              | £      | s. | d. | £       | s. | d. |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------|----|----|---------|----|----|
| Advertising Account                                                                                                                          |        |    |    | 332     | 1  | 4  |
| Brick-making Account                                                                                                                         |        |    |    | 9,724   | 4  | 4  |
| Bridge Account [Bridges in number 63]                                                                                                        |        |    |    | 99,065  | 11 | 9  |
| Charge for Direction                                                                                                                         |        |    |    | 1,911   | 0  | 0  |
| Charge for Fencing                                                                                                                           |        |    |    | 10,202  | 16 | 5  |
| Cart Establishment                                                                                                                           |        |    |    | 461     | 6  | 3  |
| Chat Moss Account                                                                                                                            |        |    |    | 27,719  | 11 | 10 |
| Cuttings and Embankments                                                                                                                     |        |    |    | 199,763 | 8  | 0  |
| Carrying Department, comprising—Amount expended in Land and Buildings for Stations and Depôts, Warehouses, Offices, &c. at the Liverpool end | 35,538 | 0  | 0  |         |    |    |
| Expended at the Manchester Station                                                                                                           | 6,159  | 0  | 0  |         |    |    |
| Side Tunnel, being the approach to the Crown-street Station                                                                                  | 2,485  | 0  | 0  |         |    |    |
| Gas-light Account, including cost of Pipes, Gasometer, &c.                                                                                   | 1,046  | 0  | 0  |         |    |    |

|                                                                                                   |        |          |    |    |
|---------------------------------------------------------------------------------------------------|--------|----------|----|----|
| Engines, Coaches, Machines, &c.                                                                   | 10,991 | 11       | 4  |    |
|                                                                                                   |        | 56,219   | 11 | 4  |
| Formation of Road                                                                                 |        | 20,568   | 15 | 5  |
| Iron Rail Account                                                                                 |        | 67,912   | 0  | 2  |
| Interest Account (balance)                                                                        |        | 3,629    | 16 | 7  |
| Land Account                                                                                      |        | 95,305   | 8  | 8  |
| Office Establishment                                                                              |        | 4,929    | 8  | 5  |
| Parliamentary and Law Expenditure                                                                 |        | 28,465   | 6  | 11 |
| Stone Blocks and Sleepers                                                                         |        | 20,520   | 14 | 5  |
| Surveying Account                                                                                 |        | 19,829   | 8  | 7  |
| Travelling Expenses                                                                               |        | 1,423    | 1  | 5  |
| Tunnel Account                                                                                    |        | 34,791   | 4  | 9  |
| Tunnel Compensation Account                                                                       |        | 9,977    | 5  | 7  |
| Waggon Account                                                                                    |        | 24,185   | 5  | 7  |
| Sundry Payments for Timber, Iron, Petty Disbursements, &c. not included in the foregoing Accounts |        | 2,227    | 17 | 3  |
|                                                                                                   |        | £739,165 | 5  | 0  |

*Extracts from the “Observations” of Mr. Booth.*

“CUTTINGS AND EMBANKMENTS.—The Excavations consist of about 722,000 cubic yards of rock and shale, and about 2,006,000 cubic yards of marle, earth, and sand. This aggregate mass has been removed to various distances, from a few furlongs to between three and four miles; and no inconsiderable portion of it has been hoisted up by machinery, from a depth of thirty to fifty feet, to be deposited on the surface above, either to remain in permanent spoil banks, as at Kenyon, or to be afterwards carried to the next embankment, as at the deep rock cutting through Olive Mount; the process in this latter case being rendered expedient from considerations of increased expedition. Where land for the deposit of spoil banks has been

purchased, the cost of the land forms part of the expenditure under this head, and a good deal of substantial and lofty walling in the deep cuttings is also included.”

“IRON RAIL ACCOUNT.—This expenditure comprises the following items:—

|                                                                                                                                                                                                                                                                                   | £.      | s. | d. |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----|----|
| Rails for a double way from Liverpool to Manchester, with occasional lines of communication, and additional side lines at the different Dépôts, being about thirty-five miles of double way,=3847 tons, at prices averaging something less than 12 <i>l.</i> 10 <i>s.</i> per ton | 48,000  | 0  | 0  |
| Cast-iron Chairs, 1428 tons, at an average of 10 <i>l.</i> 10 <i>s.</i>                                                                                                                                                                                                           | 15,000  | 0  | 0  |
| Cost of Spikes and Keys to fasten the Chairs to the Blocks, and the Rails to the Chair                                                                                                                                                                                            | 3,830   | 0  | 0  |
| Cost of Oak Plugs for the Blocks                                                                                                                                                                                                                                                  | 615     | 0  | 0  |
| Sundry Freights, Cartages, &c. &c.                                                                                                                                                                                                                                                | 467     | 0  | 2  |
|                                                                                                                                                                                                                                                                                   | £67,912 | 0  | 2  |

LAND.—This is a heavy item of expenditure. The price of land in the vicinity of large towns is usually high; and the outlay was further enhanced by numerous claims for compensation, owing to the prejudice which a few years since existed against Railways, and especially against what now appears their peculiar recommendation—the Locomotive Engine. A great change has taken place in this respect. At the close of 1828, the charge under this head was nearly 102,000*l.*, but a portion of this amount being for the dépôts, has been transferred to the carrying department.”

“The Directors, in their Report, dated the 25th of March last, estimated the total expenditure, including Warehouses, Machinery and Carriages, at 820,000*l.*, which may be apportioned as follows:—

|  | £. | s. | d. |
|--|----|----|----|
|--|----|----|----|

|                                                                                                                                                            |          |    |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----|---|
| Expenditure, as above, in actual payments, to the 31st of May [1830]                                                                                       | 739,165  | 5  | 0 |
| Outstanding engagements to the same date                                                                                                                   | 7,500    | 0  | 0 |
| For Walling the Slopes in sundry places, and completing permanent Road                                                                                     | 6,750    | 0  | 0 |
| For completing the Bridges, including the Irwell, 6000 <i>l.</i> , and Parapets of the Sankey Viaduct 1400 <i>l.</i> , and compensation in lieu of Bridges | 9,500    | 0  | 0 |
| Additional Engines, Waggon, and Machinery, part under contract for delivery                                                                                | 17,000   | 0  | 0 |
| Completing Stations, Wharfs, Warehouses, Offices, &c.                                                                                                      | 25,000   | 0  | 0 |
| Fencing at sundry places                                                                                                                                   | 3,000    | 0  | 0 |
| Contingencies                                                                                                                                              | 12,084   | 15 | 0 |
| [Making total cost of Railway]                                                                                                                             | £820,000 | 0  | 0 |

## No. II.

*The Speech of the Rev. JOHN CORRIE, F.R.S., taken from the Report of the Town's Meeting, held in Birmingham, March 4th, 1831, in support of Railways.*

The Rev. J. Corrie said, that having been desired to introduce the business of the Meeting, he had undertaken the task, from a conviction that it would be no difficult thing to show the superiority of railways to all the established modes of communication, and the benefits which the town of Birmingham in particular would derive from that great national line, which this Meeting had assembled to support. It was quite unnecessary to dwell upon the importance of safe, easy, and especially cheap means of intercourse. The whole history of human civilization attested it. Why did population first spread along the banks of great rivers? Why were the banks of the Euphrates and the Nile the seats of the earliest civilization? the cradles of science and the arts of life? Principally, if not entirely, from the facilities they afforded of cheap and easy communication. What raised

Phœnicia to such high celebrity among ancient nations? Not a narrow strip of rocky territory, but its position on the sea shore; not its abundant harvests, but its memorable ports, its Tyre and Sidon. The name and language of the Greeks were most widely diffused; their numbers, their civilization, their progress in the sciences, in the elegant and useful arts, were most marked, when in the days of the Ptolemies they possessed the first mercantile city of the world—Alexandria,—and had colonies on every little island and rocky promontory in the whole circuit of the Mediterranean. Or if we descend to modern times, What raised Italy to its early commercial eminence but the length of its indented coast? What raised great cities on the banks of the Elbe, or the Rhine, and carried manufactures into the heart of Germany, but those great rivers? What raised Flanders,—especially what raised Holland, but its canals and sea-ports? What in more recent times has occasioned the rapid extension of people, wealth, and property through the wide circuit of the United States, but the extent of its navigable rivers; those rivers on which the transcendent utility of steam navigation was fully established? Or to come nearer home, What would England be without her extent of coast, without her facility of carriage? There are many still in the vigour of their faculties who have witnessed the progress of this country during the memorable reign of George III.—a period of rapid progress in numbers, in wealth, in power, and in all the arts, with which no other period will bear one moment's comparison. How much of this was owing to our turnpike-roads—to our canals? What would Birmingham, in particular, be without them? What would be the value of the mines of Staffordshire, in the very centre of the island, and without a navigable river? But the art and industry of man has surmounted every difficulty. Birmingham is not only connected with all the kingdom by a system of excellent roads, but has a water-communication with London, with Liverpool, with Bristol, with Hull; and it is from the opening of these canals that you must date its great prosperity. In 1760, Birmingham had less than 30,000 inhabitants: by the last census it had 106,000; it has now probably 130,000. Connected with the ocean, it is connected with the world; its name is known wherever the name of England is known; its useful and ingenious productions are spread over the civilized and the uncivilized regions of the globe. But how is it that roads,—how is it that canals, have produced such wonderful effects? Solely by affording the means of easy, of safe, of regular, of rapid, of economical

communication. Which of your productions could have been sent over Europe, which to America, to India, to China, loaded with the weight of land-carriage from hence to the nearest of our ports? At least how few could have been either sent or sold! But how superior in all the particulars I have mentioned are railways to canals or turnpike roads! The ease, the smoothness, is that of one smooth bar of iron revolving on another; the motion of the best built carriage, on the best constructed road, is comparatively rough and jarring. You move with no more fatigue than from sitting in your chamber. Then for regularity,—here are no interruptions from drought, none from frost, none from snow, none from tedious periodical repairs. In safety,—inquire, and you find in the last six months upwards of 100,000 persons have passed between Liverpool and Manchester without an accident. Can you say the same of any equal length of turnpike-road? As for rapidity, this is truly astonishing,—this sets all competition at defiance; this almost realises the extravagant demand of the poet, that the gods would annihilate both space and time. It will condense the whole population of the kingdom, as to all purposes of intercourse for business or amusement; while it spreads it over the whole surface for purposes of healthful air and beautiful country scenery; and we know not yet its maximum rapidity; but our gardens, our farms, above all, our mines and manufactories, will be placed as near the great sea-ports and the sea as if Birmingham was only 30 or 40 miles from Liverpool or London. But the great point of all is the economy: and here, while we claim the right to assert a great economy, we may readily allow that there may not have been sufficient time to estimate all the expenses with which railways and engines of rapid motion may be attended. But would the low prices have been fixed between Liverpool and Manchester if the managers of this interesting establishment had not found they could with justice to the proprietors take those reduced prices? The repair of engines appears great: average this on the quantity of work performed, and you find the charge per passenger or per ton is a mere trifle,—a fraction of a farthing. We will however admit that more time is necessary to ascertain the exact expense. But we must not forget that no great improvement is ever perfected at once: whatever be the expense at first, may we not justly expect it will be reduced?—A curious paper has been published on the power, or what is called the duty, of the steam-engine in Cornwall. You know that instrument is there used most extensively to drain the deep and valuable mines. At the earliest record in

the document to which I am referring, the quantity of water raised by a bushel of coals is denoted by the number 5. The attention of the greatest engineers was drawn to this county. The celebrated Smeaton made considerable improvements; the memorable discovery which placed our late illustrious neighbour Mr. Watt in the very first rank of benefactors to his country and mankind, raised the efficiency of a single bushel to 25; subsequently, I believe, still higher. But since his time, merely by a patient attention to every minute circumstance in the construction and the working of the engine, without any new discovery, the duty has been raised from 25 to 85 or 95. Here is a progress, during no very long period: and even were the expense of the locomotive engine ten or twenty times greater than it is, there would be no reason to despair of its economy. Am I not justified then in contending that this town is most deeply interested in the establishment of a railway to Liverpool and to London? It never was more deeply interested in any contrivance to diminish the expense of carriage than at this moment. What is the state of the iron trade of Staffordshire? What the prospect of that industrious population which depends upon it? A reduction of one half in the expense of carriage, which the railways appear justly to promise, would convert a bad trade into a good one; would give Staffordshire a superiority over all competitors. At this moment the remission of the duty on sea-borne coals will narrow the market of the Staffordshire coal trade. The railway may not only restore the balance, but incline it in favour of Staffordshire, whose mines lie nearest of any to the metropolis. But the superiority of railways is a question of fact; it must be decided by actual experiment; and it has been decided by experiment between Manchester and Liverpool.

But it is said, the railway may be a good thing for the country; but what will it be for those who consume their capitals in forming it; what does it promise to its proprietors? In the first place, I would observe that the common expression,—it may be good for the country and bad for the proprietors,—expresses what never can be true: it never can be for the good of the country that any capital should be so invested as not to make an adequate return. This is a matter of calculation. Numberless railways have been constructed, (though the application of the locomotive engine is a comparatively recent improvement); and it must be possible to form some judgment of the capital required. It is true that in all great works this is

difficult, and we are often reminded of the original estimate and ultimate expense of the Liverpool and Manchester railway. But are railways the only works in which estimates have been erroneous? Has it never happened in buildings, in roads, in bridges, in canals? And is it impossible for any care, and any caution, to avoid gross mistakes? Be assured that no care will be spared on this subject by those who are inviting the public to invest a capital of three millions; their calculations are founded, not on the estimate, but on the actual expenditure of the Liverpool and Manchester railway: they suppose theirs *will cost* as much as the other actually *has cost*. Is that an objectionable basis to build upon? They pledge themselves to ascertain the probable expense, and will not rest on the judgment of any one man, however superior they may justly and confidently think him.

But a suspicion is whispered abroad, that the public are ill informed respecting the actual expenses, the profits, and the prospects of that railway, on which we ground all our plans and expectations; the statements circulated are said not to be open, candid, and above-board. Sir, I should be ashamed to enter on a justification of men so well known to the world, of such high honour, such unsullied integrity as those gentlemen, who are best known among the projectors and supporters of the Liverpool and Manchester railway. But if any one should still cherish his suspicions,—what has been their conduct? If they have deceived the public, why did they continue to hold their shares after they had reached 100*l.* per cent. premium? When they began to decline,—and they have declined,—why did they still hold them? Why are they among the first to encourage similar undertakings in favourable situations? Why were they among the earliest, why are they among the greatest shareholders in those railways, which are the object of our present meeting? But they are enthusiasts; they are blinded by their pre-conceived opinions, or by some feeling of honour. But are there not men at Liverpool, not connected with them, and able to expose their misstatements, to unmask their artifices? Many;—and how are these disclosing their sentiments respecting railways? *By embarking in a new railway, carried very nearly through the same line of country with that now at work*; for such is that projected between Liverpool and Leeds.

With regard to the traffic on the lines connected with Birmingham, I am not prepared on this occasion to go into detail; other persons well informed on this subject will address you. But look for a moment at the canals between

Liverpool and London, the Duke of Bridgewater's, the Trent and Mersey, the Staffordshire and Worcestershire, the great Birmingham canal, the Warwick, the Napton, the Coventry, the Oxford, the Grand Junction,—all of which have paid the proprietors amply, some enormously. They sufficiently indicate the extent of traffic. Add to this the land-conveyance by coaches, chaises, private carriages, waggons, vans, amounting at least to 300,000*l.* per annum, between Birmingham and London;—add the cattle sent from Leicestershire, Northamptonshire, &c. to the London markets, and which the railways will convey for less than the value of the flesh lost in driving;—add those kinds of agricultural produce which require a smooth and rapid transport. In short, statements have been made out by persons of no mean capacity, which show a profit so ample as to render the reader incredulous from its magnitude. There probably may be error, but after rigid correction the return must be ample.

Railways will no doubt be extended; and if they are what they profess to be, the sooner the better: they will be extended wherever the traffic and intercourse are great; and if we were to attempt to estimate the sum now paid for carriage on all those great lines,—I know not what it is, what it is supposed to be;—but be it a million, be it a million and a half, or two millions, if we are rejoiced at the remission of a tax of four or five hundred thousand per annum,—will the country derive no benefit from being relieved from a tax of half a million or a million on the expenses of conveyance?

The railways, which we are now considering, resting with one extremity on the metropolis, extending thence by way of Birmingham, through Staffordshire to Liverpool, and by Liverpool connected with all the north of Ireland and west of Scotland, to the north of Birmingham sending off a branch through Manchester to Leeds and the West Riding of Yorkshire,—will connect all the greatest towns, all the greatest manufacturing districts with each other and with the metropolis; and on the other hand, will thus connect the metropolis with them; they will serve as the means of intercourse among a population, which was of FOUR MILLIONS, when the whole kingdom contained less than TWELVE, and this without any wide deviation from the most direct course. The distance by Birmingham is not more than three miles more than the distance in a straight line from London

to Liverpool, not more than nine miles more than the direct distance between Manchester and London.

[He had much more to say, but had already trespassed so long upon their patience, that he would leave other topics to the many other well-informed gentlemen who were prepared to address them.]

THE END.

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## FOOTNOTES.

[\[21\]](#) Grain consumed in 1814:—

|                        | Quarters.  |
|------------------------|------------|
| Man                    | 18,750,000 |
| Animals                | 11,829,000 |
| Beer and spirits       | 4,250,000  |
| Manufacturing purposes | 171,000    |
| Making a total of      | 35,000,000 |

\*\*\* END OF THE PROJECT GUTENBERG EBOOK REMARKS ON  
THE PROPOSED RAILWAY BETWEEN BIRMINGHAM AND  
LONDON \*\*\*

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